

Concurrency Control Protocol

Database Management Systems — Gate Smashers Quick Revision & Exam Notes

1. What & Why: Aim of Concurrency Control

Core Objective

A Concurrency Control Protocol regulates the execution of concurrent transactions to guarantee **Serializability** and **Recoverability**, ensuring total **Database Consistency**.

Concurrent Transactions



Concurrency Protocol



Serializability +
Recoverability



Consistency

Problems Prevented (Why We Need It)

- **Dirty Read:** Reading uncommitted data
- **Unrepeatable Read:** Data changed mid-transaction
- **Lost Update:** Overwriting concurrent writes
- **Incorrect Summary:** Partial aggregate calculation

2. Types of Protocols & Lock Basics

Protocols include *Simple Locking*, *Shared-Exclusive Locking*, *2PL (Strict/Rigorous)*, and *Timestamp Protocols*. The exam core centers on **Shared-Exclusive Locking**.

Shared Lock — S Lock

Purpose: Read-Only Access

Used when a transaction only needs to read a data item without modifying it.

```
S(A) // Shared Lock on A
READ(A)
U(A) // Unlock A
```

Exclusive Lock — X Lock

Purpose: Read + Write Access

Required when a transaction intends to perform both read and write operations.

```
X(A) // Exclusive Lock on A
READ(A)
WRITE(A)
U(A) // Unlock A
```

3. Shared (S) vs Exclusive (X) Summary

Lock Type	Symbol	Meaning	READ	WRITE	Memory Trick
Shared Lock	S(A)	Shared Access	YES	NO	S = See → Read Only
Exclusive Lock	X(A)	Exclusive Access	YES	YES	X = eXclusive → Read + Write

🔥 **Lock Selection Rule:** Only READ → Use S Lock | READ + WRITE → Use X Lock

4. Lock Compatibility Matrix (CRITICAL FOR EXAMS)

Determines if a newly requested lock on data item A can be granted when another lock is already held.

Already Granted	Requested Lock	
	S	X
S	YES	NO
X	NO	NO

🧠 Golden Compatibility Trick

Only S-S is Compatible!

- **S - S = YES** (Read-Read doesn't conflict)
- **S - X = NO** (Read vs Write conflict)
- **X - S = NO** (Write vs Read conflict)
- **X - X = NO** (Write vs Write conflict)

5. Case-by-Case Breakdown

1. S → S (Granted): Both T1 & T2 only read. No Read-Read conflict exists; multiple shared locks are permitted.

2. S → X (Denied): T1 is reading, T2 wants to write. Prevents Read-Write conflict.

3. X → S (Denied): T1 has exclusive write rights. Denied to prevent Write-Read conflict for T2.

4. X → X (Denied): T1 and T2 both want write access. Denied to prevent concurrent Write-Write conflict.

6. Common Exam Traps

✗ **Trap 1:** "An S lock guarantees exclusive access to a data item."

→ **FALSE.** Other transactions can also acquire an S lock concurrently on the same item.

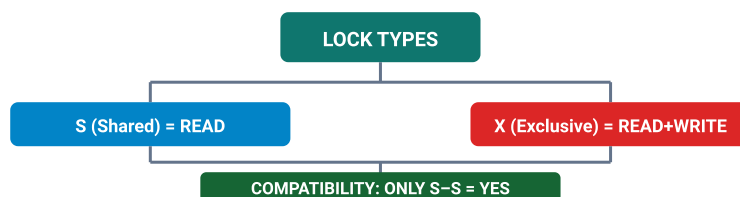
✗ **Trap 2:** "An X lock allows ONLY write operations."

→ **FALSE.** An X lock allows BOTH READ and WRITE operations.

✗ **Trap 3:** "S-X or X-S locks can coexist if operations don't overlap."

→ **FALSE.** Compatibility fails instantly if an X lock is involved with any other lock.

7. 30-Second Final Revision Map



🏆 **Golden Formula:** S = Read | X = Read + Write | Compatibility = S-S Only